

Power BI Foundations & Data Blending

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1. Power BI Foundations & Interface

What is Power BI?

Power BI is a Microsoft business intelligence tool that helps organizations turn raw data into meaningful reports and interactive dashboards.

With Power BI, we can:

- Connect to Excel, CSV, databases and other data sources
- Clean and transform data
- Combine multiple datasets
- Create relationships between tables
- Perform calculations
- Create charts and dashboards
- Filter and interact with reports
- Publish and share reports

Power BI Workflow

Connect → Clean → Model → Visualize → Interact → Share

Or:

```
graph TD; A[Data Sources] --> B[Power Query]; B --> C[Data Model]; C --> D[Visualizations]; D --> E[Report]; E --> F[Publish / Share];
```

Data Sources
↓
Power Query
↓
Data Model
↓
Visualizations
↓
Report
↓
Publish / Share

Power BI Components

Power BI Desktop

The main development application.

Used for:

- Data loading
- Data cleaning
- Data modelling
- Creating reports
- Creating calculations
- Designing visuals

Power BI Service

The online/cloud environment used to:

- Publish reports
- Share reports
- Collaborate
- Create dashboards
- Manage published content

Power BI Mobile

Used to access Power BI reports from mobile devices.

Power BI Desktop Interface

When Power BI Desktop is open, introduce the major areas.

1. Report View

This is where we create the report pages and visualizations.

Use it for:

- Charts
- Cards
- Tables
- Slicers
- Maps
- Report layout

2. Data/Table View

Used to inspect the data inside loaded tables.

You can:

- View rows
- Examine columns
- Check data types
- Understand the structure of the data

3. Model View

Used to see how different tables are connected.

This is particularly important when working with multiple tables.

4. Visualizations Pane

Used to:

- Select a visual
- Add fields
- Format the visual
- Configure visual properties

5. Data/Fields Pane

Contains:

- Tables
- Columns
- Measures

Practical: Load an Excel Dataset

Use a simple `Sales.xlsx` file containing:

Order Date	Product	Category	Customer	Region	Sales	Quantity	Profit
01-Jan-26	Laptop	Technology	ABC	South	50000	2	8000
05-Jan-26	Chair	Furniture	XYZ	West	20000	5	4000
10-Feb-26	Monitor	Technology	ABC	South	30000	3	5000

Steps

1. Open Power BI Desktop.
2. Select **Home**.
3. Select **Get Data**.
4. Choose **Excel workbook**.
5. Browse to `Sales.xlsx`.
6. Select the required sheet/table.
7. Click **Load**.

Now the table will appear in Power BI.

Question

"Looking at this dataset, what business questions can we answer?"

Expected answers:

- What are total sales?
- Which category has the highest sales?
- Which region performs best?
- Which products generate the most profit?
- How are sales changing over time?
- Which customers are generating the most revenue?

2. Graph & Chart Types

The important thing is not simply how to create a chart, but **when to use each chart**.

A. Column / Bar Chart

Purpose

Use column or bar charts when comparing categories.

Example:

Category	Sales
Furniture	₹250K
Technology	₹420K
Office Supplies	₹180K

The business question is:

Which category has higher sales?

Steps to Create a Column Chart

1. Go to **Report View**.
2. Select **Clustered column chart**.
3. Drag **Category** into the **X-axis**.
4. Drag **Sales** into the **Y-axis / Values**.
5. Resize the visual.
6. Select the **Format** options.
7. Add a suitable title such as **Sales by Category**.

Demonstration

Ask:

"Which category is generating the highest sales?"

B. Line Chart

Purpose

Use line charts when you want to understand trends over time.

Example:

- January → ₹100K
- February → ₹120K
- March → ₹145K
- April → ₹160K

Steps

1. Select **Line chart**.
2. Drag **Order Date** to the **X-axis**.
3. Drag **Sales** to the **Y-axis**.
4. Change the date hierarchy if required.
5. Use **Month/Year** for a monthly trend.
6. Give the visual a title: **Monthly Sales Trend**.

Ask

- Are sales increasing?
 - Which month had the highest sales?
 - Is there a downward trend?
-

C. Pie / Donut Chart

Purpose

Use these charts to show part-to-whole relationships.

For example:

- Technology – 45%
- Furniture – 30%
- Office Supplies – 25%

Steps

1. Select **Donut chart**.
2. Drag **Category** to **Legend**.
3. Drag **Sales** to **Values**.
4. Format data labels.
5. Add a title.

Tip

"Pie charts work best when there are only a few categories. If you have many categories, use a bar chart instead."

D. Card

Purpose

A card displays one important number.

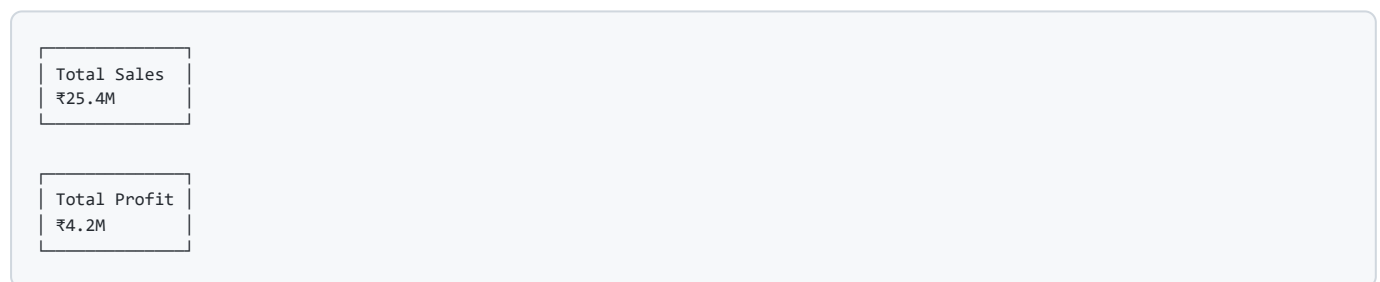
Examples:

- Total Sales
- Total Profit
- Total Orders
- Customer Count

Steps

1. Select **Card**.
2. Drag **Sales** into the card.
3. Power BI automatically aggregates the values.
4. Format the number.
5. Add a title such as **Total Sales**.

You can create multiple cards:



E. Table

Purpose

Tables are useful when users need to see exact values.

Steps

1. Select **Table**.
2. Drag **Product**.
3. Drag **Sales**.
4. Drag **Profit**.
5. Arrange the columns as required.

Example:

Product	Sales	Profit
Laptop	₹5L	₹80K
Monitor	₹3L	₹50K

F. Matrix

A Matrix is similar to a Pivot Table and is useful for hierarchical analysis.

Example:

Region ↓ State ↓ City

Steps

1. Select **Matrix**.
2. Add **Region** to **Rows**.
3. Add **State** below **Region**.
4. Add **Sales** to **Values**.
5. Use the expand/collapse buttons to navigate the hierarchy.

G. Scatter Chart

Purpose

Used to investigate relationships between numerical variables.

Example: Sales vs Profit

Steps

1. Select **Scatter chart**.
2. Add **Sales** to **X-axis**.
3. Add **Profit** to **Y-axis**.
4. Add **Category** to **Legend** if appropriate.
5. Add **Product** to **Details** if appropriate.

“Do products with higher sales also generate higher profit?”

H. Treemap

Useful for displaying hierarchical part-to-whole relationships.

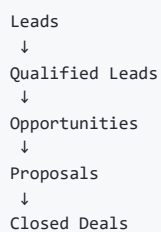
Steps

1. Select **Treemap**.
 2. Add **Category** to **Category**.
 3. Add **Product** to **Details**.
 4. Add **Sales** to **Values**.
-

I. Funnel

Useful for processes containing multiple stages.

Example:



```
graph TD; Leads --> Qualified_Leads[Qualified Leads]; Qualified_Leads --> Opportunities; Opportunities --> Proposals; Proposals --> Closed_Deals[Closed Deals];
```

Leads
↓
Qualified Leads
↓
Opportunities
↓
Proposals
↓
Closed Deals

Steps

1. Select **Funnel chart**.
 2. Put **Stage** in **Category**.
 3. Put **Count** or **Sales** in **Values**.
-

J. Gauge

Used to compare actual performance against a target.

Example:

- Actual Sales = ₹8M
- Target Sales = ₹10M

Steps

1. Select **Gauge**.
2. Add **Sales** to **Value**.
3. Add **Target Sales** to **Target value**.
4. Configure minimum/maximum values if necessary.

Easy Chart Selection Rule

Business Requirement	Recommended Visual
Compare categories	Bar / Column
Show trend	Line
Show proportion	Pie / Donut
Show KPI	Card
Show exact data	Table
Show hierarchy	Matrix
Show relationship	Scatter
Show hierarchy + proportion	Treemap
Show process stages	Funnel
Actual vs Target	Gauge

3. Geographic Mapping

Power BI can convert geographical data into maps.

Common geographical fields:

- Country
- State
- City
- Postal Code
- Latitude
- Longitude

Example:

Country	State	City	Sales
India	Tamil Nadu	Chennai	₹5M
India	Karnataka	Bengaluru	₹7M
India	Maharashtra	Mumbai	₹8M

Creating a Map

Depending on the Power BI version and enabled mapping visuals, the available map visual may differ. Use the map visual available in the **Visualizations** pane.

Steps

1. Load your geographical dataset.
2. Go to **Report View**.
3. Select the appropriate **Map** visual.
4. Add **City** or **State** to the **Location** field.
5. Add **Sales** to the size/value field.
6. Power BI will plot the locations.
7. Format the map as required.

Important: Data Category

Sometimes Power BI may not correctly understand whether a field represents a city, state or country.

For example: Chennai — Power BI needs to know that this is a **City**.

Steps

1. Go to **Data/Table View**.
2. Select the geographical column.
3. Go to **Column tools**.
4. Find **Data category**.
5. Select the appropriate category:
 - City
 - State or Province
 - Country/Region
 - Postal Code
 - Latitude
 - Longitude

This helps Power BI interpret geographical information correctly.

Latitude & Longitude

Latitude and longitude provide more precise locations.

Example — Chennai:

- Latitude: 13.0827
- Longitude: 80.2707

Steps

1. Add the map visual.
2. Add **Latitude** to the latitude field.
3. Add **Longitude** to the longitude field.
4. Add **Sales** as the value/size.
5. Add **City** or another identifying field if required.

Mapping Exercise

To create:

- **Sales by State**

Then create a slicer for:

- **Category**

Students select:

Technology

“What happens to the map when we select Technology?”

This naturally introduces dashboard interactivity.

4. Data Joining / Relationships

This is one of the most important Power BI concepts.

Instead of storing everything in one huge table, analytical models often use multiple related tables.

Example Data Model

Sales

OrderID	ProductID	CustomerID	Sales
1001	P01	C01	5000
1002	P02	C02	3000

Product

ProductID	Product	Category
P01	Laptop	Technology
P02	Chair	Furniture

Customer

CustomerID	Customer	Region
C01	ABC Ltd	South
C02	XYZ Ltd	West

The common fields are:

```
Sales[ProductID]
  ↓
Product[ProductID]
```

and:

```
Sales[CustomerID]
  ↓
Customer[CustomerID]
```

What is a Relationship?

A relationship connects tables through a common field/key.

For example:

```
Product
ProductID
  1
  |
  |
  *
Sales
ProductID
```

One product can appear in many sales transactions.

Primary Key

A primary key uniquely identifies each record.

Example:

ProductID
P01
P02
P03

There should be **no duplicate ProductIDs** in the Product table.

Foreign Key

A foreign key refers to the key in another table.

Example — Sales[ProductID]:

ProductID
P01
P01
P02
P03
P01

Duplicates are normal because the same product can be sold many times.

Relationship Types

One-to-Many

Most commonly used.

Product 1 — * Sales

One product → many sales transactions.

Many-to-One

The same relationship viewed from the opposite direction.

One-to-One

One record corresponds to one record.

Many-to-Many

Multiple records in one table can correspond to multiple records in another.

“Many-to-many relationships should be designed carefully because they can cause ambiguity and unexpected results.”

Creating Relationships in Power BI

Method 1 — Model View

1. Load all tables.
2. Select **Model View**.
3. Locate the **Sales** table.
4. Locate the **Product** table.
5. Drag: `Sales[ProductID]` onto: `Product[ProductID]` .
6. Power BI opens the relationship configuration.
7. Check:
 - Cardinality
 - Cross-filter direction
 - Related tables
8. Click **OK**.

Repeat for Customer:

```
Sales[CustomerID]
      ↓
Customer[CustomerID]
```

Why Relationships Matter

Incorrect relationships can cause:

- Incorrect totals
- Duplicate results
- Filters not working correctly
- Unexpected visual behaviour
- Incorrect calculations

Practical Exercise

- Sales table
- Products table
- Customers table

Ask is to:

1. Load all tables.
2. Open **Model View**.
3. Identify common keys.
4. Create relationships.
5. Create **Sales by Category**.
6. Create **Sales by Region**.
7. Add a **Region** slicer.
8. Test whether the visuals respond correctly.

5. Data Blending / Combining Data

In Power BI, data can be combined in different ways.

The two concepts must clearly understand are: **Append** and **Merge**.

Append

Append means:

■ Add rows from one table underneath another table.

Example:

January Sales	February Sales
Row 1	Row 3
Row 2	Row 4
↓ APPEND	
All Sales	
Row 1	
Row 2	
Row 3	
Row 4	

Use **Append** when tables have similar structures.

Append Example

Sales_January

Date	Product	Sales
Jan 1	Laptop	5000
Jan 2	Chair	3000

Sales_February

Date	Product	Sales
Feb 1	Laptop	6000
Feb 2	Chair	3500

We can append them into Sales_All:

Date	Product	Sales
Jan 1	Laptop	5000
Jan 2	Chair	3000
Feb 1	Laptop	6000
Feb 2	Chair	3500

How to Append in Power Query

1. Select **Home** → **Transform data**.
2. Power Query Editor opens.
3. Select **Home** → **Append Queries**.
4. Choose:
 - Two tables, or
 - Three or more tables.
5. Select the tables.
6. Click **OK**.
7. Review the combined data.
8. Select **Close & Apply**.

Easy memory trick:

Append = Add Rows

Merge

Merge means:

Use a matching key to bring columns from another table.

Example:

Sales

ProductID	Sales
P01	5000
P02	3000

Product

ProductID	Category
P01	Technology
P02	Furniture

Merge allows us to bring **Category** into Sales.

Result:

ProductID	Sales	Category
P01	5000	Technology
P02	3000	Furniture

How to Merge in Power Query

1. Select **Home** → **Transform data**.
2. Select the first table.
3. Go to **Home** → **Merge Queries**.
4. Select the second table.
5. Select the matching column in the first table.
6. Select the matching column in the second table.
7. Select the required **Join Kind**.
8. Click **OK**.
9. A new column containing the matched table appears.
10. Click the **expand** button.
11. Select the columns you want to bring into the first table.
12. Click **OK**.
13. Select **Close & Apply**.

Easy memory trick:

Merge = Add Columns

Append vs Merge

Append	Merge
Adds rows	Adds columns
Vertical combination	Horizontal combination
Similar table structures	Matching key required
Union-like operation	Join-like operation

Merge / Join Types

Introduce the common join types.

Join Type	Description
Left Outer	Keep everything from the first table and matching records from the second
Right Outer	Keep everything from the second table and matching records from the first
Inner	Keep only records that match in both tables
Full Outer	Keep records from both tables
Anti Join	Useful for finding unmatched records

Example:

Which products exist in the Product Master but have never been sold?

6. Data Cleaning & Transformation

Raw data is rarely ready for analysis.

Typical problems include:

- Blank rows
- Duplicate records
- Incorrect data types
- Missing values
- Extra spaces
- Spelling differences
- Incorrect dates
- Unnecessary columns
- Multiple pieces of information in one column

Power Query is the main environment used to clean and transform this data.

Open Power Query

Steps

1. Go to **Home**.
2. Select **Transform data**.
3. Power Query Editor opens.
4. Perform the required transformations.
5. Select **Close & Apply** when finished.

A. Remove Duplicates

Example:

Customer
ABC
ABC
XYZ

Steps

1. Select the relevant column.
2. Go to **Home** → **Remove Rows** → **Remove Duplicates**.

Use this carefully—don't remove duplicates from a transaction table merely because values repeat. Repeated transactions may be valid.

B. Remove Blank Rows

Steps

1. Select **Home**.
2. Select **Remove Rows**.
3. Select **Remove Blank Rows**.

C. Handle Null Values

Example:

Product	Profit
Laptop	50,000
Monitor	null
Chair	10,000

A null does not automatically mean zero.

It could mean:

- Data missing
- Not available
- Not applicable
- Not yet calculated

To understand the **business meaning** before replacing nulls.

D. Change Data Types

Correct data types are essential.

Column	Correct Type
Sales	Decimal Number
Quantity	Whole Number
Order Date	Date
Customer ID	Text

Steps

1. Open Power Query.
2. Select the column.
3. Select the **Data Type** icon/dropdown.
4. Choose the appropriate type.

E. Trim Extra Spaces

Example:

- " Chennai "
- "Chennai"

These may appear identical but can behave differently.

Steps

1. Select the text column.
2. Go to **Transform**.
3. Select **Format**.
4. Select **Trim**.

For non-printing characters, use **Clean** where appropriate.

F. Split Columns

Example:

Full Name
John Smith

We want:

First Name	Last Name
John	Smith

Steps

1. Select **Full Name**.
2. Go to **Transform** → **Split Column**.
3. Choose **By Delimiter**.
4. Select **Space**.
5. Choose how to split.
6. Click **OK**.

G. Replace Values

Suppose we have:

- TN
- Tamil Nadu
- Tamilnadu

We want a standardized value: **Tamil Nadu**

Steps

1. Select the column.
2. Go to **Transform** → **Replace Values**.
3. Enter the old value.
4. Enter the replacement value.
5. Click **OK**.

H. Rename Columns

Instead of:

- Col1
- Col2
- Col3

Use meaningful names:

- Customer Name
- Order Date
- Sales Amount

Steps: Double-click the column name or right-click and select **Rename**.

I. Remove Unnecessary Columns

Remove columns that aren't required for:

- Analysis
- Relationships
- Calculations
- Reporting

Steps

1. Select the unnecessary column.
2. Right-click.
3. Select **Remove**.

7. Dashboard Interactivity

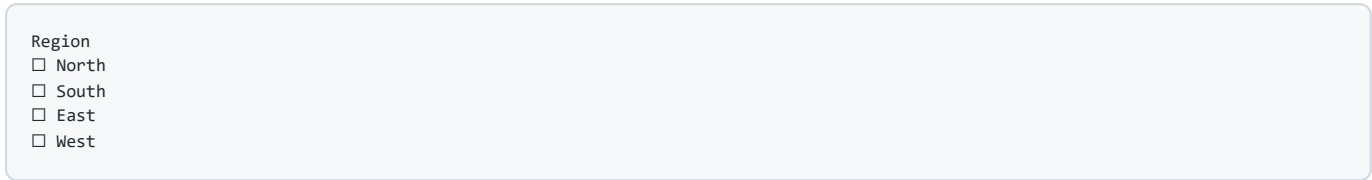
The key concept:

A Power BI report is interactive. Users should be able to explore the data rather than simply look at static charts.

A. Slicers

Slicers allow users to filter report data interactively.

Example:



Region
☐ North
☐ South
☐ East
☐ West

Create a Slicer

1. Go to **Report View**.
2. Select **Slicer**.
3. Drag **Region** into the slicer's field.
4. Resize the slicer.
5. Select a region.
6. Observe the other visuals.

Create additional slicers for:

- Region
- Category
- Product
- Date

B. Visual Interactions

Suppose the page contains:

- Sales by Category
- Sales by Region
- Profit by Product

When you click **Technology** in Sales by Category, the other visuals may change.

Demonstrate Interaction

1. Select the first visual.
2. Go to **Format**.
3. Select **Edit interactions**.
4. Other visuals will display interaction controls.
5. Choose whether each visual should:
 - **Filter**
 - **Highlight**
 - **Do nothing**

Explain the difference between filtering and highlighting.

C. Drill Down

Drill-down allows users to move through a hierarchy.

Example:

Year
↓
Quarter
↓
Month
↓
Day

or:

Country
↓
State
↓
City

Steps

1. Create a visual.
2. Add multiple hierarchy levels to the axis.
3. Use the drill controls displayed on the visual.
4. Select **Drill Down**.
5. Move from Year → Quarter → Month.

D. Drill-through

Drill-through takes the user to another report page containing details about a selected item.

Example:

Page 1 — Sales Dashboard

User selects: **Laptop**

Page 2 — Product Details

Show:

- Sales
- Profit
- Orders
- Customers
- Monthly trend

Basic Steps

1. Create a new report page.
2. Add the field you want to drill through on.
3. Place that field in the page's **Drillthrough** area.
4. Design the detailed page.
5. Return to the main report.
6. Right-click a relevant data point.
7. Select **Drill through**.
8. Select the detail page.

E. Tooltips

Tooltips provide additional information when the user hovers over a visual.

Example:

Product: Laptop
Sales: ₹5.2M
Profit: ₹850K
Orders: 1,250

Basic Approach

1. Select a visual.
2. Add relevant fields to the visual's **Tooltips** field well, where available.
3. Hover over the data point.
4. Power BI displays the additional information.

F. Bookmarks

Bookmarks save a particular report state.

They can be used for:

- Navigation
- Show/hide panels
- Switching between views
- Interactive storytelling

Basic Steps

1. Open **View**.
2. Enable the **Bookmarks** pane.
3. Configure the report state.
4. Select **Add** in Bookmarks.
5. Rename the bookmark.
6. Use buttons to navigate between bookmarks if required.

8. Final Hands-on Exercise

Given small Sales dataset.

Ask to build a **Mini Sales Dashboard**.

Requirement 1 — KPI Cards

Create:

- Total Sales
- Total Profit
- Total Orders

Requirement 2 — Column Chart

Create: **Sales by Category**

Field	Placement
Category	Axis
Sales	Values

Requirement 3 — Line Chart

Create: **Monthly Sales Trend**

Field	Placement
Order Date	Axis
Sales	Values

Requirement 4 — Map

Create: **Sales by State**

Field	Placement
State	Location
Sales	Size/Value

Requirement 5 — Region Slicer

Create **Region** as a slicer.

Requirement 6 — Category Filter

Create a slicer/filter for **Category**.

Requirement 7 — Interaction

Students should verify:

When I select a category, the relevant cards, charts and map respond.

Suggested Final Dashboard Layout

